

Create:-

In SQL create statement is used to create new database objects such as tables, view, indexes or databases. The most common usage is created which defines table structure

i, Database:-

Create a database it helps to store all land records

Create database LandManagement;

To use the base we use the following Command

use LandManagement;

ii, Create table:-

This is used to create a new table statement in SQL is used to create new table in database & we add columns & constraints.

```
create table land-owners (owner_id int primary key,  
                           first_name varchar(50) not null,  
                           second_name varchar(50) not null,  
                           phone_No varchar(10) Unique);
```

```
create table land (land_id int primary key, owner_id int,  
                  Area decimal(10,2), area varchar(20);  
                  Zone varchar(20), status varchar(20);  
                  value decimal(15,2), Registration_date date,  
                  Foreign key (owner_id) references  
                  land-owners (owner_id));
```

View:-

It is a virtual table & it doesn't occupy any physical memory
It displays data from one or more tables

Create view available-land as

select land-id, area, value from land
where status = 'Available';

Index:-

It is used to improve speed of data retrieval operations

Create index zone on land(zone-type);

Copy table:-

It creates empty backup table

Create table archived-land like land;

Copy table data:-

insert into archived-land select * from land;

It inserts data into backup table

ALTER:-

It is used to modify the structure of an existing table, such as adding, deleting or changing columns.

Add column:-

It adds new column into the table

Alter table land add property-tax decimal(10,2);

modify column:-

It changes the name or data type or data size

Alter table land modify column zone varchar(50);

Drop column:-

it drops a column from the table

alter table land drop property-tax;

Constraint:-

It is a rule applied to a table column in a database to restrict the type of data that can be stored & ensure data accuracy & integrity.

alter table land add check_value check (value > 100000);

alter table land add uni-land-zone unique (land-id, zone);

Insert:-

It inserts values into the table

insert into land_owners values (36451, 'Dogiparthi', 'Sai Naga Piyosh',
'9492528135');

insert into land values (6521, 36451, 1500.20, 'Commercial', 'Available',
'1500000.30', '2026-06-06');

Update:-

It updates the inserted values

update land set value = '1600000.30' where land_id = 6521;

Delete:-

It deletes the values

delete from land where land_id = 6520;

Truncate:-

Empties the table but it keeps the structure

truncate table archived-land;

Show:-

It shows all the tables or databases created

Show tables;

Select:-

It is used to retrieve data from one or more tables in a database

Where:-

Used to filter records that meet a specified condition

Operators:-

Symbols used to perform comparisons or calculations in SQL queries.

Limit:-

Used to restrict the number of rows returned in a query result

Select * from land where (Status = 'Available' and area > 10000) or not zone = 'Agriculture';

Like:-

It is used in where clause to search for specified pattern in a column

Select * from land-owners where last-name like 'B%';

Between:-

It is used in where clause to search the range

Select * from land where market between 1000000 and 5000000;

IN / NOT IN:-

IN - used to select records where a column value matches any value in list

Not IN - used to select records where column value don't match any value in list

Select * from land where zone in ('Commercial', 'Agriculture');

Distinct :-

used to remove duplicates value & return only unique records from query

Order by :-

used to sort the result of a query based on one or more columns.

Desc/asc :-

used with order by to sort data in descending/ascending order

Limit :-

used to restrict n:o of rows returned in a query result

select distinct zone from land order by zone desc limit 5;

Exists / not exists :-

Returns true/false if subquery returns at least one/no records.

select first_name from land-owners o where exists (select 1 from land p where p.owner_id = o.owner_id);

Group By :-

used to group rows with the same values in specified columns, often with aggregate functions.

Having :-

used to filter groups of records after the group by clause

AS :-

used to assign as alias to a column or table in a query

select zone, count(land-id) as totalLand from land group by zone
having count(land-id) > 10;

Arithmetic Operators:-

Used to perform mathematical calculations (eg:- +, -, *, /)

Logical operators:-

Used to combine multiple conditions in a query (eg:- AND, OR, NOT)

Relational operators:-

Used to compare two values (eg:- =, >, <, <=, >=, !=);

Select land_id, (market_value * 1.05) as next-projected-value from land;

Aggregate Function:-

Count:- Returns n:o of rows or values in column

Sum:- Returns total sum of numeric values in column

Avg:- Returns average value of numeric column

Max:- Returns the large value in a column

min:- Returns the small value in a column

Select count(*) as total, sum (area) as tot-area, Avg (value) as avg-value, max (value) as high-value, min (value) as low-value from land;

String & text functions:-

These are SQL functions used to manipulate, modify & analyze text or character data in a database.

Select concat (First_name, ' ' Last_name) as full_name, upper(status) from land_owners join land on land_owners.owner_id = Land.owner_id;

Date functions:-

select land-id, year(Registration-date) as reg-year from land;

Mathematical & statistical functions:-

These are used to perform calculations & statistical operations on numeric data.

select round(area, 0) as rounded-area, sqrt(area) as sqr-area, stddev(value) as marked-value from land;

JOINS:-

These are used in SQL to combine rows from two or more tables based on related column.

select o.first-name, p.parce-id, p.area from land o

Join .

land P on o.owner-id = p.~~Owner~~-id;

Set operations & set Comparison:-

set operations:- used to combine the results of two or more select queries (eg:- UNION, UNION ALL, INTERSECT, EXCEPT)

Set Comparison:- used to compare a value with a set of values returned by a subquery (eg:- Any, All, Some)

UNION:-

Combines the results of both queries but removes duplicate rows

select owner-id from land where zone = 'Residential'

Union

select owner-id from land where zone = 'Commercial';

Union All:-

Combines results but keeps all duplicates. it is faster than Union
select owner-id from land where zone = 'Residential'.

Union all

select owner-id from land where zone = 'Commercial';

Intersect:-

It returns only the rows that appear in result of both queries

select owner-id from land where zone = 'Residential'

intersect

select owner-id from land where zone = 'Commercial';

Except:-

It returns the rows from the first query that do not appear
in square secondary query

select owner-id from land where zone = 'Residential'

Except

select owner-id from land where zone = 'Commercial';

Any:-

this operator evaluates to true if the condition is satisfied for at least one of the values returned by subquery

select land_id, zone, area from land where area > any (select area from land where zone = 'commercial');

Some:-

It is simply alias for any. It works same way & are completely interchangeable.

select land_id, zone, value from land where value = some (select value from land where zone = 'Agriculture');

All:-

This evaluates to true only if the condition is satisfied for every single value returned by subquery

select land_id, zone, value from land where value > All (select value from land where zone = 'Agriculture');

Subqueries:-

It is a query written inside another SQL query that provides data to the main query.

select * from land where value > (select avg(value) from land);

select land_id, zone, area from land where area > (select avg(area) from land);

Multi-Row Subquery:-

In this we use operators like IN, Not IN, Any, All
Select owner_id, first_name, last_name, phone_no from land_owners
where owner_id in (select owner_id from land where
zone = 'Commercial');

Correlated Subquery:-

It depends on data from outer query to run. Because of this, the inner query must execute once for every single row evaluated by outer query.

Select land_id, zone, value from land l1 where value >
(select avg(value) from land l2 where l1.zone = l2.zone);

Subquery in select clause:-

It list to calculate a separate value for each row.

select first_name, last_name (select count(land_id) from land
where land.owner_id = land.owner_id) as total_land_owned
from land_owners;